

Derivations/Proofs you need to know

Afleidings/Bewyse wat jy moet ken

- Derive the equations of motion for constant acceleration from the definitions of average acceleration and average velocity. (ch 2) using both differentiation and integration.

Herlei die vergelykings vir beweging met konstante versnelling van die definisies van gemiddelde versnelling en gemiddelde snelheid. (hfst 2) deur differensiasie en integrasie methods te gebruik.

- Show that the range of a projectile is given by $R = \frac{v_0^2}{g} \sin 2\theta_0$ from the basic equations of motion for constant acceleration. (ch 4)

Wys dat $R = \frac{v_0^2}{g} \sin 2\theta_0$ (die rykwydte van 'n projektiel) van die basiese vergelykings van beweging vir konstante versnelling.

- From first principles show that the acceleration during uniform circular motion is given by $a = \frac{v^2}{r}$.

Van eerste beginsels, wys dat die versnelling tydens uniforme sirkelbeweging gegee word deur $a = \frac{v^2}{r}$.

- Derive the work-kinetic energy theorem from Newton's second law $F = ma$.
Herlei die arbeid-kinetiese energies stelling vanaf Newton se tweede wet $F = ma$.

- Derive the formula for work done by a spring force.
Herlei die formule vir arbeid verrig deur 'n veerkrag.

- Show, using the basic definitions for potential and kinetic energy, that the mechanical energy E of a linear oscillator is $E = \frac{1}{2}kx^2$.
Gebruik die definisies van potensieële energie en kinetiese energie, en wys dat die meganiese energie E van 'n lineêre ossilator $E = \frac{1}{2}kx^2$ is.

- Derive the period of a simple pendulum $T = 2\pi\sqrt{\frac{L}{g}}$.

Herlei die vergelyking vir die periode van 'n eenvoudige slinger. $T = 2\pi\sqrt{\frac{L}{g}}$